

● RAG Learning

1.2 Fractions, Decimals & Percentages

AQA GCSE Mathematics 8300 — Number — spec refs N8 (fractions), N10–N12 — Papers 1 (non-calculator), 2 and 3 (calculator)

Fractions, decimals and percentages are three ways of writing the same value. You need to calculate **exactly** with fractions (no rounding), move freely between the three forms, use fractions and percentages as operators (multipliers), and work with fractions in ratio problems. At Higher tier you also convert between recurring decimals and fractions.

Quick Reference: Equivalences to Learn

Fraction	Decimal	Percentage	Fraction	Decimal	Percentage
$\frac{1}{2}$	0.5	50%	$\frac{1}{8}$	0.125	12.5%
$\frac{1}{4}$	0.25	25%	$\frac{3}{8}$	0.375	37.5%
$\frac{3}{4}$	0.75	75%	$\frac{1}{3}$	0.333...	$33\frac{1}{3}\%$
$\frac{1}{5}$	0.2	20%	$\frac{1}{20}$	0.05	5%

Know these by heart for the non-calculator paper. Also: $\frac{1}{10} = 0.1 = 10\%$, $\frac{2}{3} = 0.666\dots = 66\frac{2}{3}\%$, $\frac{1}{100} = 0.01 = 1\%$.

1.2.1 Fraction Basics

- **Proper fraction:** numerator smaller than denominator, e.g. $\frac{3}{5}$.
- **Improper fraction:** numerator at least as big as the denominator, e.g. $\frac{7}{4}$.
- **Mixed number:** a whole number and a proper fraction, e.g. $1\frac{3}{4} = \frac{7}{4}$.

- **Equivalent fractions:** have the same value, e.g. $\frac{2}{3} = \frac{8}{12}$ (multiply top and bottom by the same number).
- **Simplest form:** numerator and denominator have no common factor other than 1 — divide both by their HCF.

Mixed to improper: $a\frac{b}{c} = \frac{ac + b}{c}$, e.g. $2\frac{3}{4} = \frac{2 \times 4 + 3}{4} = \frac{11}{4}$.

Improper to mixed: divide; the remainder is the new numerator, e.g. $\frac{19}{15} = 1\frac{4}{15}$.

1.2.2 Calculating Exactly with Fractions (N8)

Adding and subtracting

Write the fractions over a **common denominator** (the LCM of the denominators is neatest), then add or subtract the numerators only.

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$$

Worked Example:

Work out $\frac{3}{4} + \frac{5}{6}$.

LCM of 4 and 6 is 12: $\frac{9}{12} + \frac{10}{12} = \frac{19}{12} = 1\frac{7}{12}$

Worked Example:

Work out $3\frac{1}{4} - 1\frac{5}{6}$.

Improper fractions: $\frac{13}{4} - \frac{11}{6} = \frac{39}{12} - \frac{22}{12} = \frac{17}{12} = 1\frac{5}{12}$

Multiplying and dividing

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$$

Always change mixed numbers to improper fractions first. To divide, multiply by the **reciprocal** ("keep, flip, multiply"). Cancel common factors before multiplying to keep numbers small.

Worked Example:

Work out $2\frac{2}{5} \times 1\frac{1}{4}$.

$\frac{12}{5} \times \frac{5}{4}$: cancel the 5s and $\frac{12}{4} = 3$. Answer: **3**

Worked Example:

Work out $2\frac{2}{3} \div 1\frac{3}{5}$.

$$\frac{8}{3} \div \frac{8}{5} = \frac{8}{3} \times \frac{5}{8} = \frac{40}{24} = \frac{5}{3} = 1\frac{2}{3}$$

Worked Example:

A rectangle measures $3\frac{1}{2}$ cm by $2\frac{2}{7}$ cm. Find its exact area and perimeter.

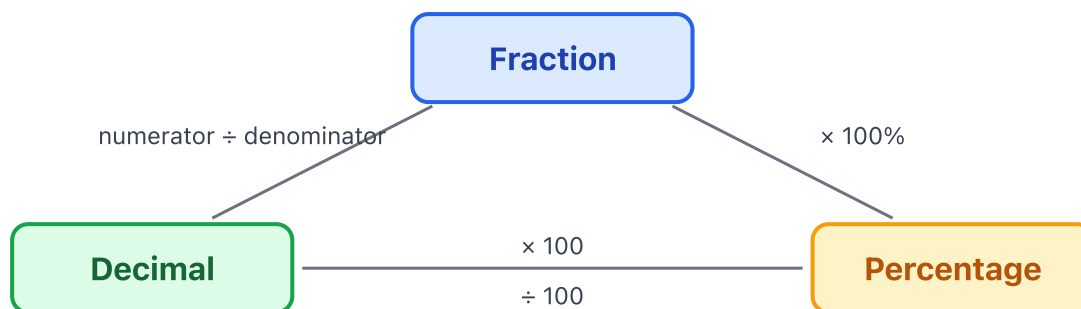
$$\text{Area} = \frac{7}{2} \times \frac{16}{7} = \frac{16}{2} = 8 \text{ cm}^2$$

$$\text{Perimeter} = 2 \left(\frac{7}{2} + \frac{16}{7} \right) = 2 \left(\frac{49}{14} + \frac{32}{14} \right) = \frac{81}{7} = 11\frac{4}{7} \text{ cm}$$

Exam Tip: "Calculate exactly" means leave answers as fractions (not rounded decimals).

Never add the denominators: $\frac{1}{4} + \frac{1}{3} \neq \frac{2}{7}$. With negative mixed numbers, convert the whole thing: $-2\frac{1}{2} = -\frac{5}{2}$, not $-2 + \frac{1}{2}$.

1.2.3 Converting and Ordering FDP (N10)



Moving between the three forms. Decimal to fraction: write over 10, 100 or 1000 and simplify.

- **Fraction → decimal:** divide the numerator by the denominator, or make an equivalent fraction over 10, 100 or 1000: $\frac{3}{25} = \frac{12}{100} = 0.12$.
- **Terminating decimal → fraction:** use place value, then simplify: $0.375 = \frac{375}{1000} = \frac{3}{8}$; $3.5 = \frac{7}{2}$.
- **Percentage:** "per cent" means out of 100: $35\% = \frac{35}{100} = \frac{7}{20} = 0.35$.

Worked Example:

Write in order, smallest first: 0.62, $\frac{5}{8}$, 63%, $\frac{3}{5}$

As decimals: 0.62, 0.625, 0.63, 0.6

Answer: $\frac{3}{5}$, 0.62, $\frac{5}{8}$, 63%

Terminating or recurring?

A fraction in its simplest form gives a **terminating** decimal only when its denominator has no prime factors other than 2 and 5. Otherwise it gives a **recurring** decimal.

Fraction	Denominator	Decimal
$\frac{7}{40}$	$2^3 \times 5$	0.175 (terminates)
$\frac{3}{22}$	2×11	0.1363636... (recurs)
$\frac{5}{6}$	2×3	0.8333... (recurs)

1.2.4 Recurring Decimals and Fractions Higher

A **recurring decimal** has a digit or block of digits that repeats for ever. Dots go over the first and last digit of the repeating block:

$$0.\dot{7} = 0.777\dots \quad 0.4\dot{5} = 0.454545\dots \quad 0.1\dot{3}\dot{6} = 0.13636\dots \quad 0.\dot{1}2\dot{3} = 0.123123\dots$$

Fraction → recurring decimal

Use short division, adding zeros after the decimal point until the remainders start to repeat.

Worked Example:

Write $\frac{7}{11}$ as a recurring decimal.

$7.0000 \div 11$: $70 \div 11 = 6 \text{ r } 4$; $40 \div 11 = 3 \text{ r } 7$; $70 \div 11 = 6 \text{ r } 4 \dots$ the remainders repeat.

$$\frac{7}{11} = 0.636363\dots = \mathbf{0.\dot{6}\dot{3}}$$

Recurring decimal → fraction

1. Let x = the recurring decimal.
2. Multiply by a power of 10 so the repeating block lines up: $\times 10$ for one repeating digit, $\times 100$ for two, $\times 1000$ for three.
3. Subtract to remove the recurring part.
4. Solve for x and simplify.

Worked Example:

Prove that $0.4\dot{5} = \frac{5}{11}$.

$$\begin{aligned}
 x &= 0.454545 \dots \\
 100x &= 45.454545 \dots \\
 99x &= 45 \\
 x &= \frac{45}{99} = \frac{5}{11}
 \end{aligned}$$

Worked Example:

Write $0.2\dot{1}$ ($= 0.2111\dots$) as a fraction.

The recurring part does not start straight after the point, so use two multiples whose recurring tails match:

$$\begin{aligned}
 100x &= 21.111 \dots \\
 10x &= 2.111 \dots \\
 90x &= 19 \\
 x &= \frac{19}{90}
 \end{aligned}$$

Worked Example:

Show that $0.1\dot{3}\dot{6} = \frac{3}{22}$.

$$\begin{aligned}
 1000x &= 136.3636 \dots \\
 10x &= 1.3636 \dots \\
 990x &= 135 \\
 x &= \frac{135}{990} = \frac{3}{22}
 \end{aligned}$$

Shortcut pattern	Example
$0.\dot{a} = \frac{a}{9}$	$0.\dot{7} = \frac{7}{9}$
$0.\dot{a}\dot{b} = \frac{ab}{99}$	$0.\dot{2}\dot{7} = \frac{27}{99} = \frac{3}{11}$
$0.\dot{a}b\dot{c} = \frac{abc}{999}$	$0.\dot{1}2\dot{3} = \frac{123}{999} = \frac{41}{333}$

Exam Tip: When the question says "prove" or "show", you must use the algebraic method ($100x - x$ etc.) — quoting the shortcut is not enough. Always write out enough digits (at least two full cycles) so it is clear the recurring parts cancel, and finish by simplifying.

1.2.5 Fractions and Percentages as Operators (N12)

"Of" means multiply. A fraction or percentage applied to an amount acts as an **operator**.

Fraction of an amount: $\frac{3}{5}$ of 40 = $40 \div 5 \times 3 = 24$

Percentage of an amount: $r\%$ of $A = \frac{r}{100} \times A$, e.g. 17.5% of £240 = $0.175 \times 240 = £42$

Increase by $r\%$: multiply by $(1 + \frac{r}{100})$ Decrease by $r\%$: multiply by $(1 - \frac{r}{100})$

Change	Multiplier
Find 15%	0.15
Increase by 8%	1.08
Increase by 12.5%	1.125
Decrease by 30%	0.7
Decrease by $\frac{3}{8}$	$\frac{5}{8}$ (or 0.625)

Worked Example:

Without a calculator, find 35% of £64.

10% = £6.40, so 30% = £19.20; 5% = £3.20. 35% = 19.20 + 3.20 = **£22.40**.

Worked Example:

A house worth £185 000 rises in value by 6%, then falls by 6%. Find its final value.

$$185\,000 \times 1.06 \times 0.94 = \textbf{£184\,334}$$

It is not back to £185 000 because the 6% fall is taken from the larger value of £196 100.

Overall multiplier $1.06 \times 0.94 = 0.9964$, a 0.36% decrease.

Worked Example:

A coat costs £72. Shop B takes 30% off, then a further 5% off the sale price. Find the price.

$$72 \times 0.7 \times 0.95 = \textbf{£47.88}$$
 (not the same as 35% off, which would be £46.80).

Exam Tip: Successive percentage changes multiply — never add them. Using a single multiplier is quicker and is the method AQA expects on calculator papers. Percentage change, reverse percentages and compound interest are covered in topic 3.2.

1.2.6 Fractions in Ratio Problems (N11)

A ratio compares parts with parts; a fraction compares a part with the **whole**.



Red : blue = 2 : 3, so red is $\frac{2}{5}$ of the total and blue is $\frac{3}{5}$. The red amount is $\frac{2}{3}$ of the blue amount.

- Ratio $a : b \rightarrow$ the first part is $\frac{a}{a+b}$ of the whole.
- If $\frac{3}{7}$ of a class are boys, then $\frac{4}{7}$ are girls and boys : girls = 3 : 4.
- "A gets $\frac{5}{8}$ of what B gets" means $A : B = 5 : 8$.

Worked Example:

Amy and Ben share £91. Amy receives $\frac{5}{8}$ of the amount Ben receives. How much does each get?

Amy : Ben = 5 : 8, so 13 parts. One part = $91 \div 13 = 7$.

Amy = $5 \times 7 = \text{£}35$, Ben = $8 \times 7 = \text{£}56$.

Worked Example:

Boys and girls in a club are in the ratio 5 : 3. $\frac{3}{5}$ of the boys and $\frac{1}{3}$ of the girls walk. What fraction of members walk?

Boys are $\frac{5}{8}$ of the club: $\frac{3}{5} \times \frac{5}{8} = \frac{3}{8}$. Girls are $\frac{3}{8}$: $\frac{1}{3} \times \frac{3}{8} = \frac{1}{8}$.

Total walking = $\frac{3}{8} + \frac{1}{8} = \frac{1}{2}$.

1.2.7 Multi-step Fraction Problems

When a fraction is taken of "the rest" or "the remainder", work out what fraction of the *original* each part represents.

Worked Example:

Tom spends $\frac{2}{5}$ of his money on rent and $\frac{1}{4}$ of the rest on food. He has £540 left. How much did he start with?

After rent, $\frac{3}{5}$ remains. He keeps $\frac{3}{4}$ of that: $\frac{3}{5} \times \frac{3}{4} = \frac{9}{20}$ of the original.

$\frac{9}{20}$ of the total = 540 $\Rightarrow \frac{1}{20} = 60 \Rightarrow \text{total} = \text{£}1200$.

Worked Example:

A tank is $\frac{3}{5}$ full. After 24 litres are used it is $\frac{1}{3}$ full. Find its capacity.

$\frac{3}{5} - \frac{1}{3} = \frac{9}{15} - \frac{5}{15} = \frac{4}{15}$ of the tank is 24 litres.

$\frac{1}{15} = 6$ litres, so capacity = **90 litres**.

Exam Tip: Draw a bar model for "fraction of the remainder" problems — split the bar into the first fraction, then split the remaining section again. Check your answer by working forwards from the total you found.

Summary

Skill	Key method
Add / subtract fractions	Common denominator, then combine numerators
Multiply / divide fractions	Improper fractions; multiply across; divide = multiply by reciprocal
FDP conversion	Fraction $\rightarrow$ decimal: divide; decimal $\rightarrow$ %: $\times 100$
Operators	"Of" means $\times$; increase by $r\%$: $\times \left(1 + \frac{r}{100}\right)$
Ratio and fractions	Part $\div$ total parts
Recurring decimals Higher	$x, 10^n x$, subtract, solve, simplify